US012385719B2

(12) **United States Patent**
Teetzel et al.

(10) **Patent No.: US 12,385,719 B2**
(45) **Date of Patent: Aug. 12, 2025**

(54) **WEAPON SLING WITH INTEGRAL CONNECTORS FOR WIRED COMMUNICATION AND/OR POWER TRANSFER**

(71) Applicant: **WILCOX INDUSTRIES CORP.**, Newington, NH (US)

(72) Inventors: **James W. Teetzel**, Portsmouth, NH (US); **Elliott S. Turner**, Newmarket, NH (US)

(73) Assignee: **Wilcox Industries Corp.**, Newington, NH (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 113 days.

(21) Appl. No.: **18/423,698**

(22) Filed: **Jan. 26, 2024**

(65) **Prior Publication Data**

US 2024/0384968 A1 Nov. 21, 2024

Related U.S. Application Data

(60) Provisional application No. 63/444,302, filed on Feb. 9, 2023.

(51) **Int. Cl.**
F41C 33/00 (2006.01)
F41C 27/00 (2006.01)

(52) **U.S. Cl.**
CPC ***F41C 33/002*** (2013.01); ***F41C 27/00*** (2013.01)

(58) **Field of Classification Search**
CPC F41C 33/00; F41C 33/002; F41C 33/001; F41C 23/02; F41C 27/00; F41C 33/005; A41D 13/0012; A41D 1/002; A45F 2005/006; A45F 5/00; A45F 2003/142; A45F 3/14; A45F 5/1533; Y10S 224/902; Y10S 224/93
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,211,321	A *	5/1993	Rodriguez	A45F 5/00 224/604
11,168,956	B2 *	11/2021	Dilling	F41C 33/002
2019/0033035	A1 *	1/2019	Echevarria	F41C 33/002
2020/0278174	A1 *	9/2020	Gill	F41H 1/02
2024/0164464	A1	5/2024	Teetzel et al.	
2024/0255257	A1	8/2024	Teetzel et al.	

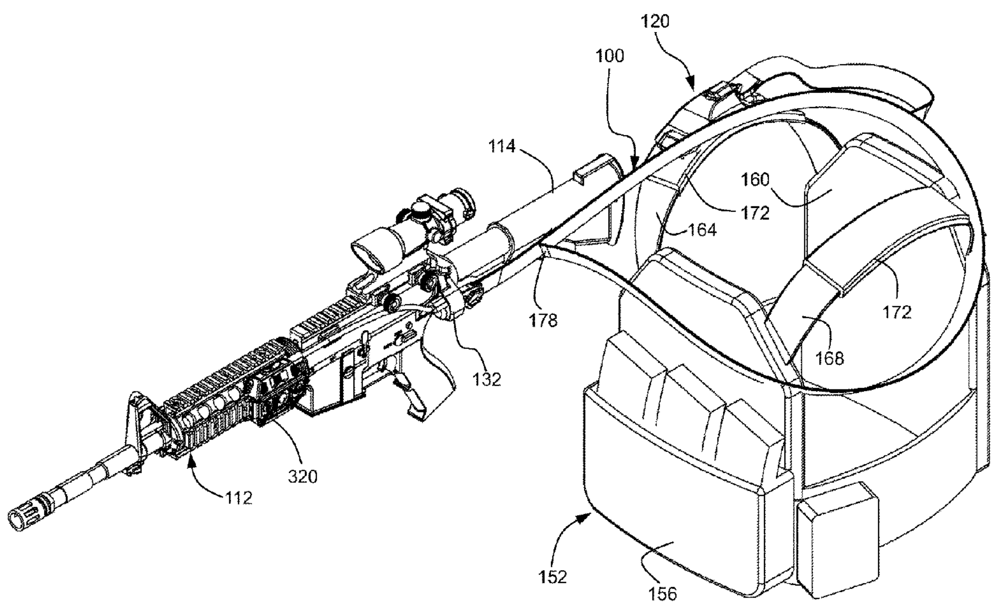
* cited by examiner

Primary Examiner — Corey N Skurdal
(74) *Attorney, Agent, or Firm* — McLane Middleton, Professional Association

(57) **ABSTRACT**

A weapon sling includes a strap having first and second free ends and defining a main branch between the first and second free ends. The strap member is formed of an elongate hollow flexible, e.g., textile, element. A loop branch attached to the main branch is configured to be looped around a user's body. First and second connector assemblies are secured to the respective first and second free ends. The first connector assembly is for detachable coupling to a weapon connector on the weapon. The second connector assembly is for detachable coupling to a hub connector on a hub worn by the user. A plurality of electrical conductors extend within the elongate hollow flexible element for electrically coupling electrical contacts on the first connector assembly with electrical contacts on the second connector assembly for providing communications between a host computer on the weapon and the hub.

19 Claims, 16 Drawing Sheets



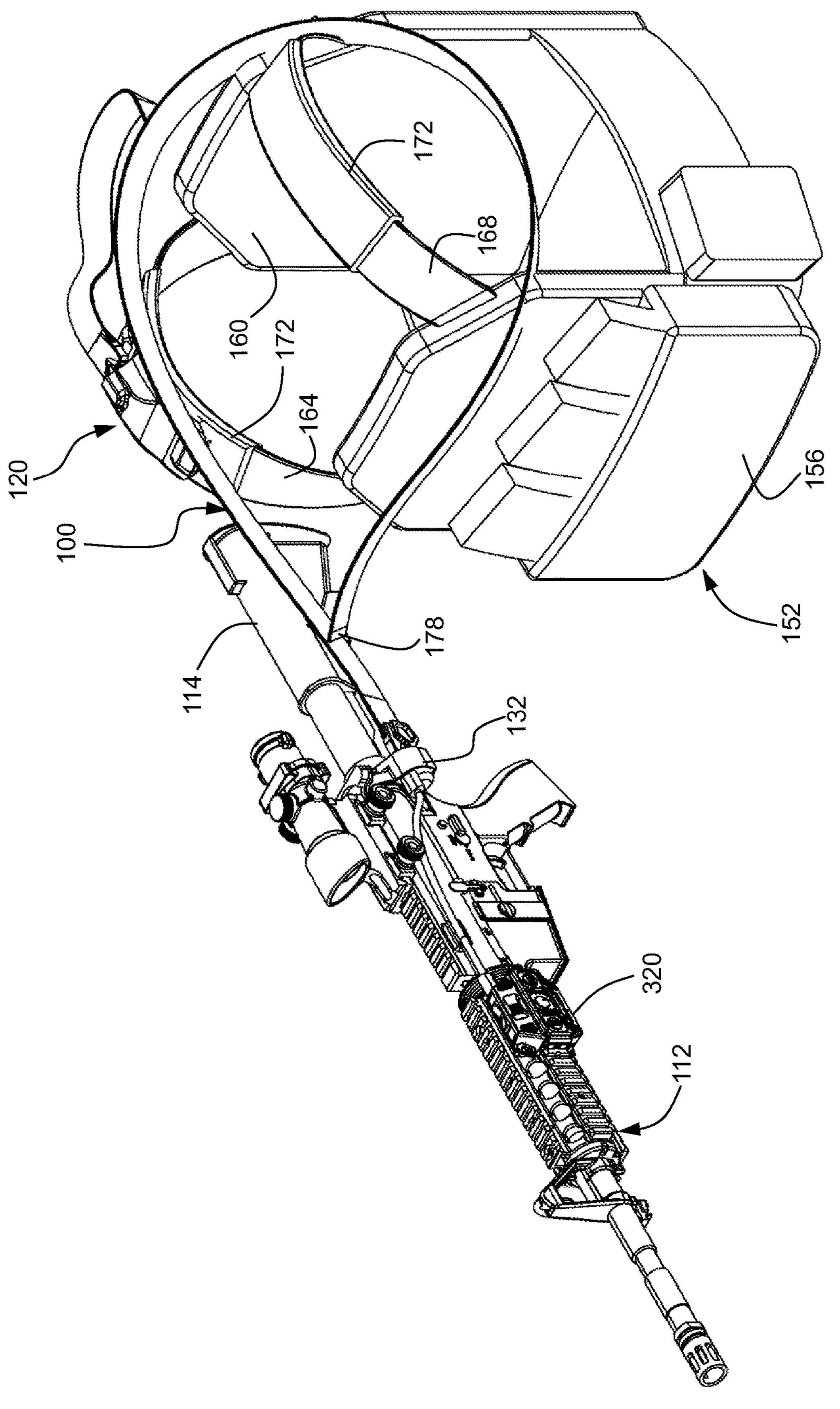


FIG. 1

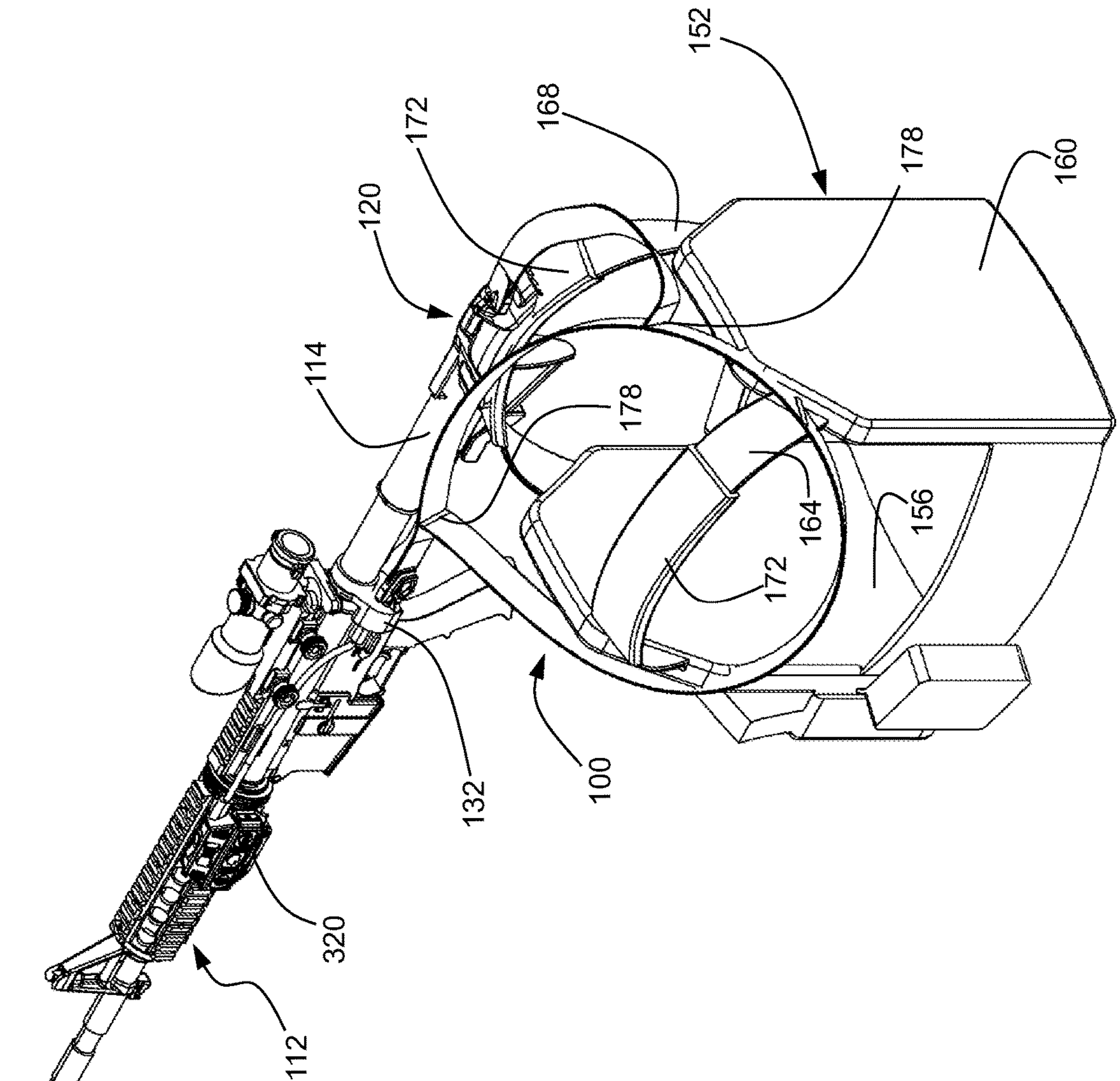


FIG. 2

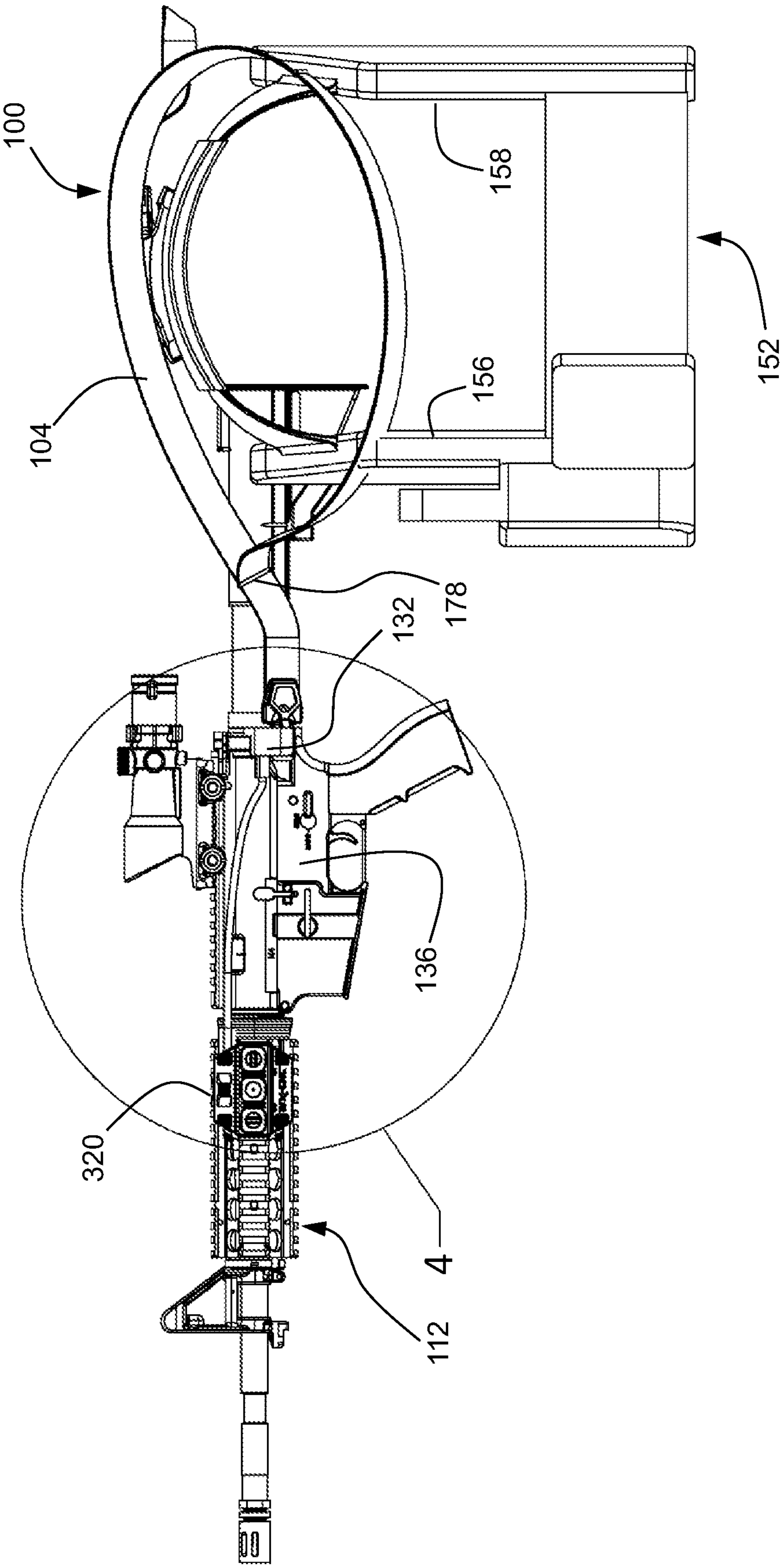


FIG. 3

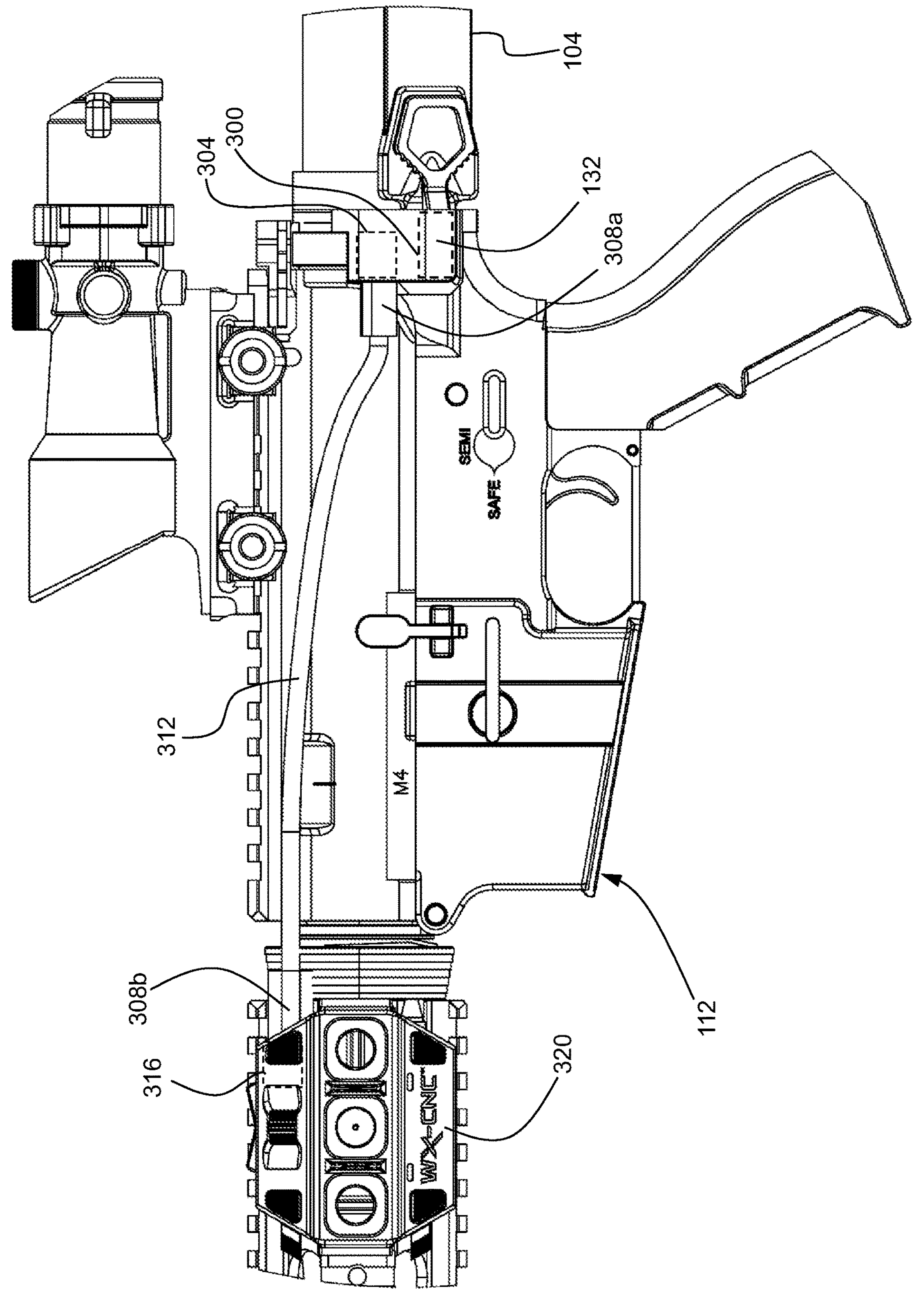


FIG. 4

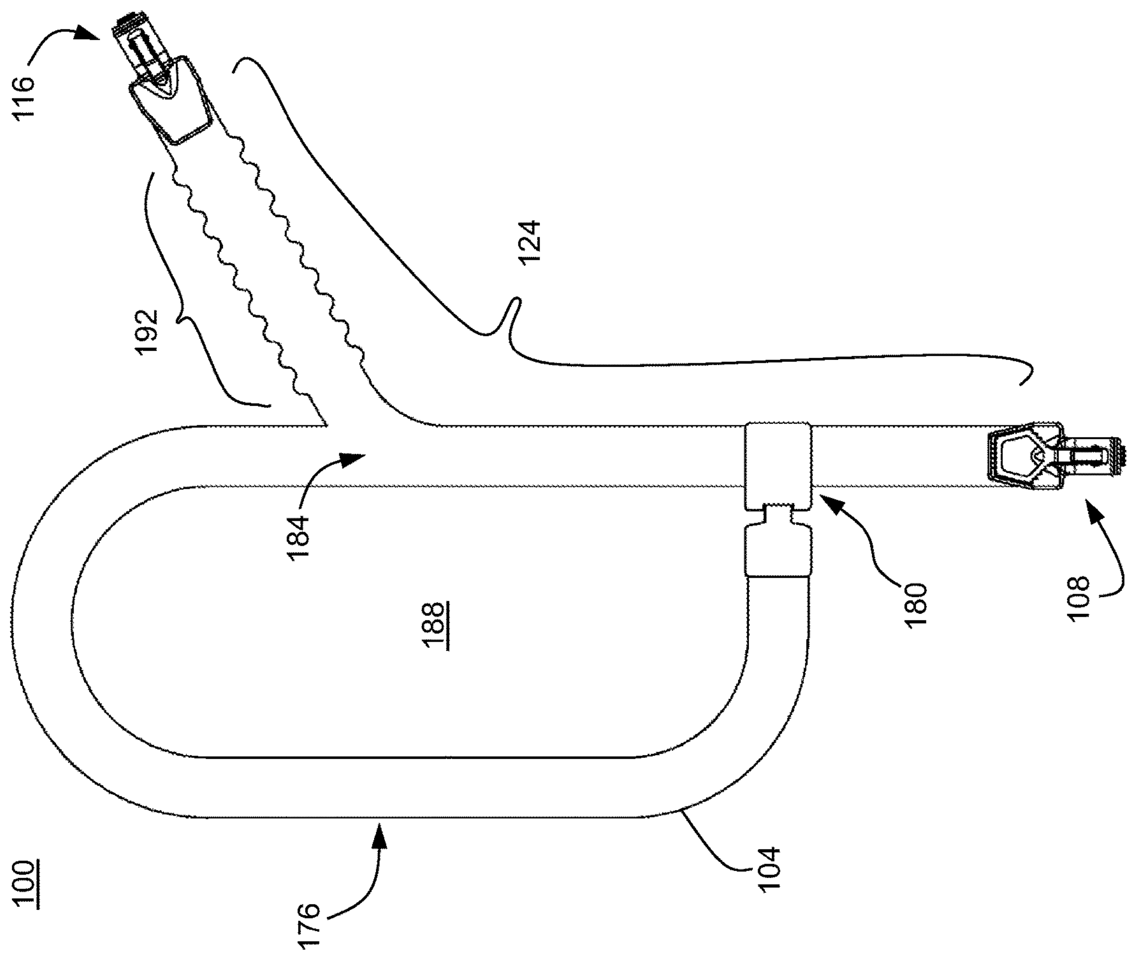


FIG. 5

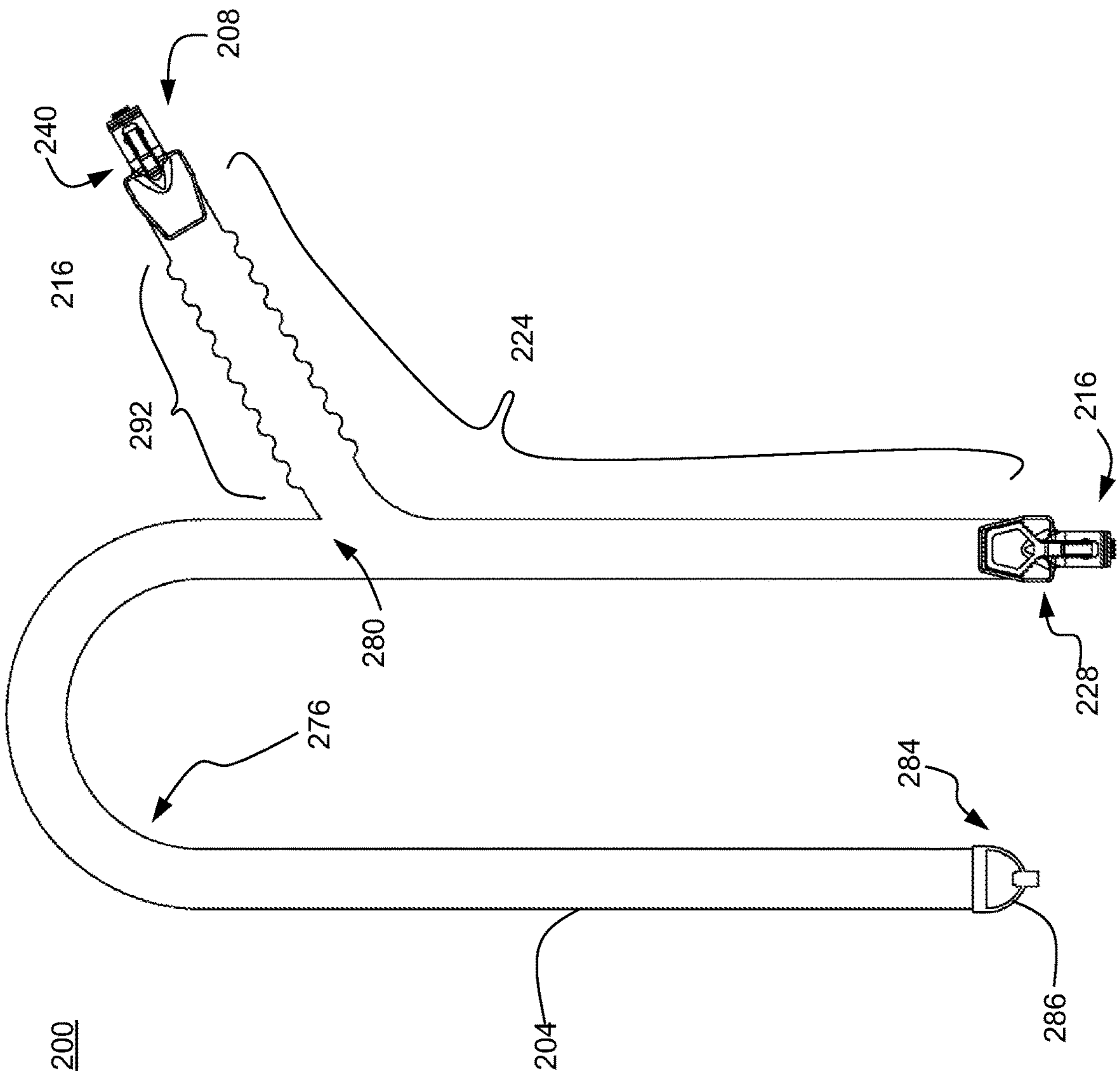


FIG. 6

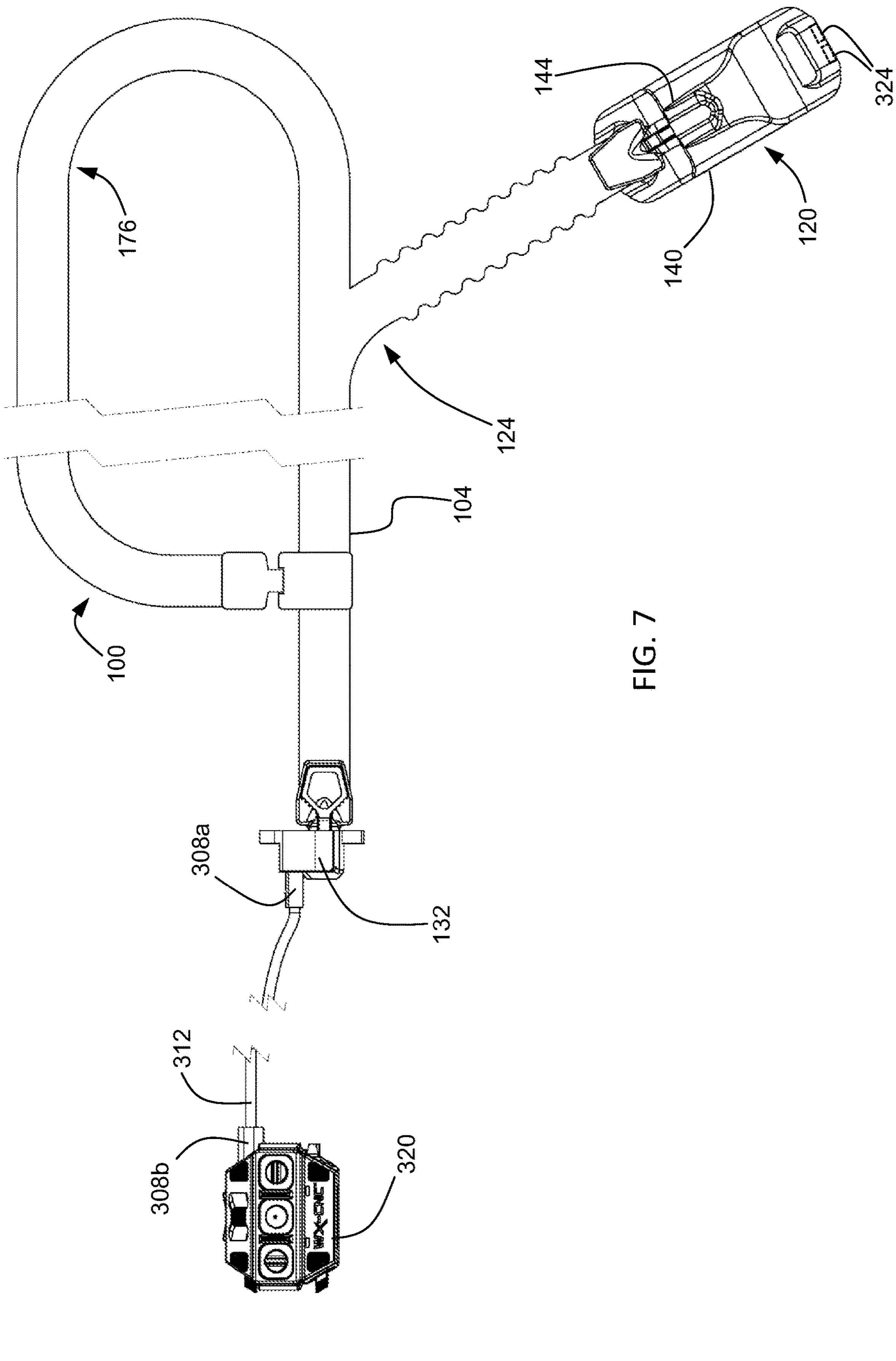


FIG. 7

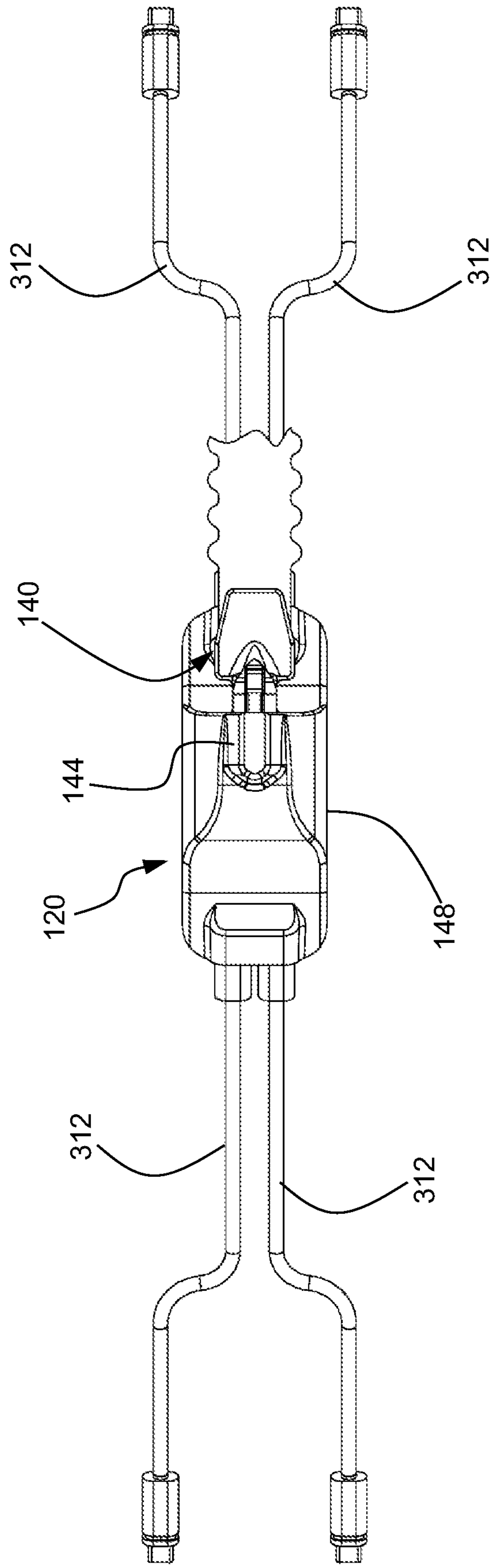


FIG. 8

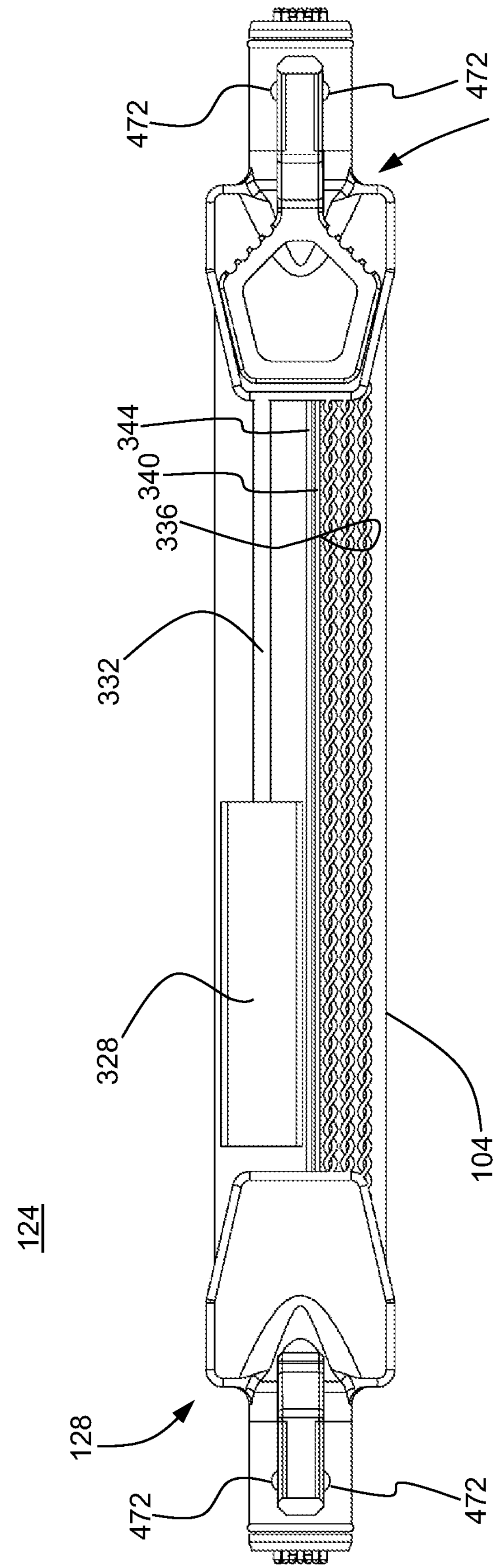


FIG. 9

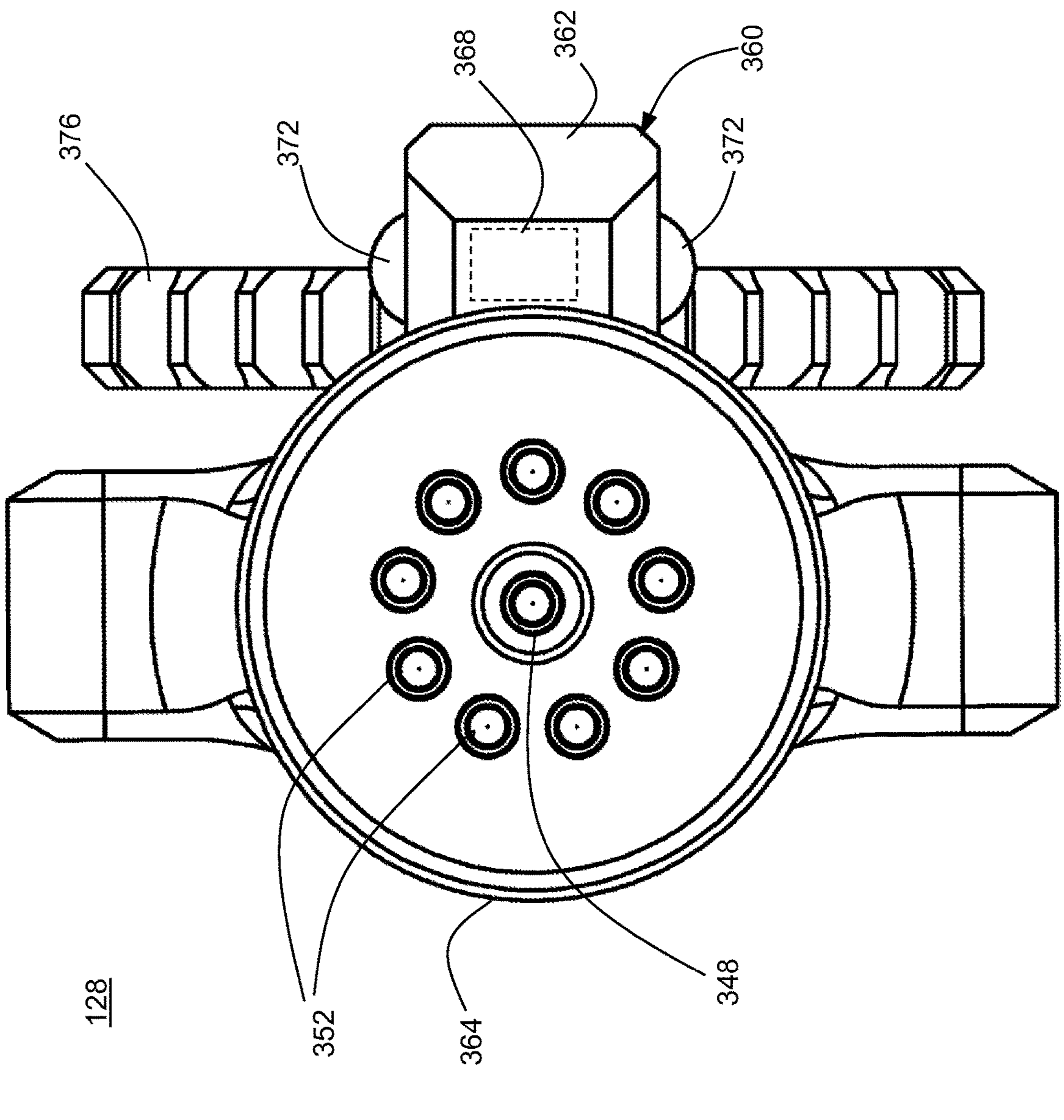


FIG. 10

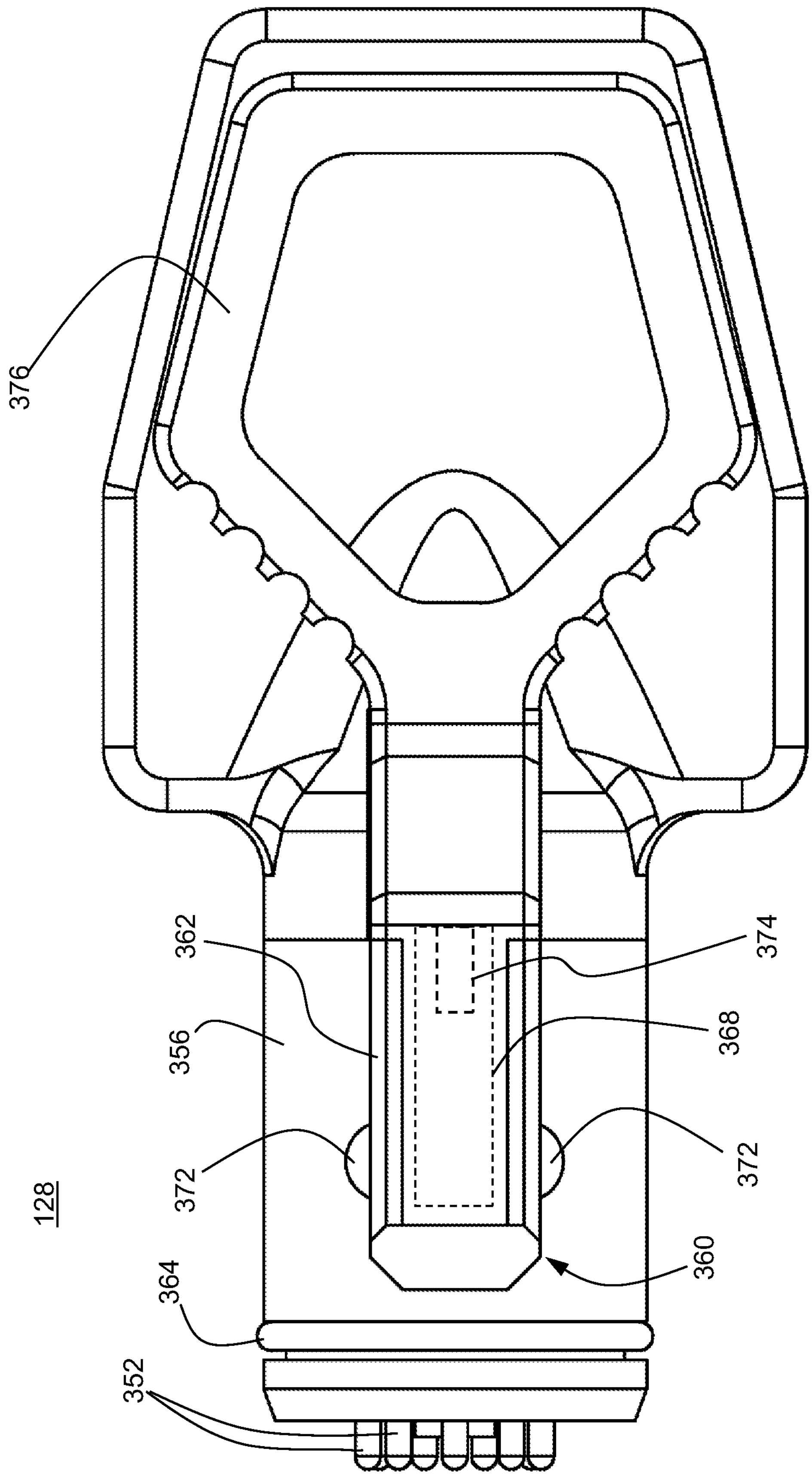


FIG. 11

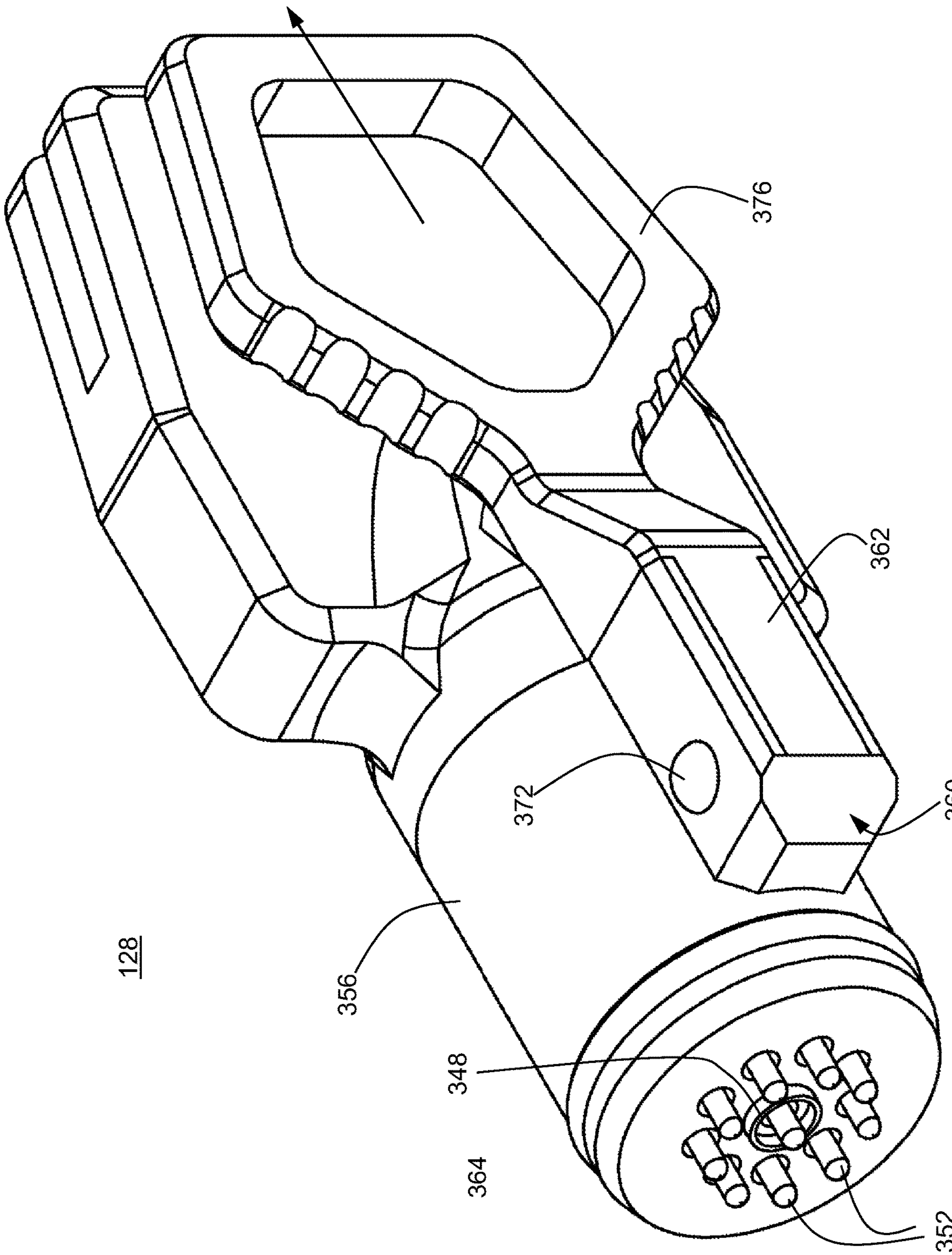


FIG. 12

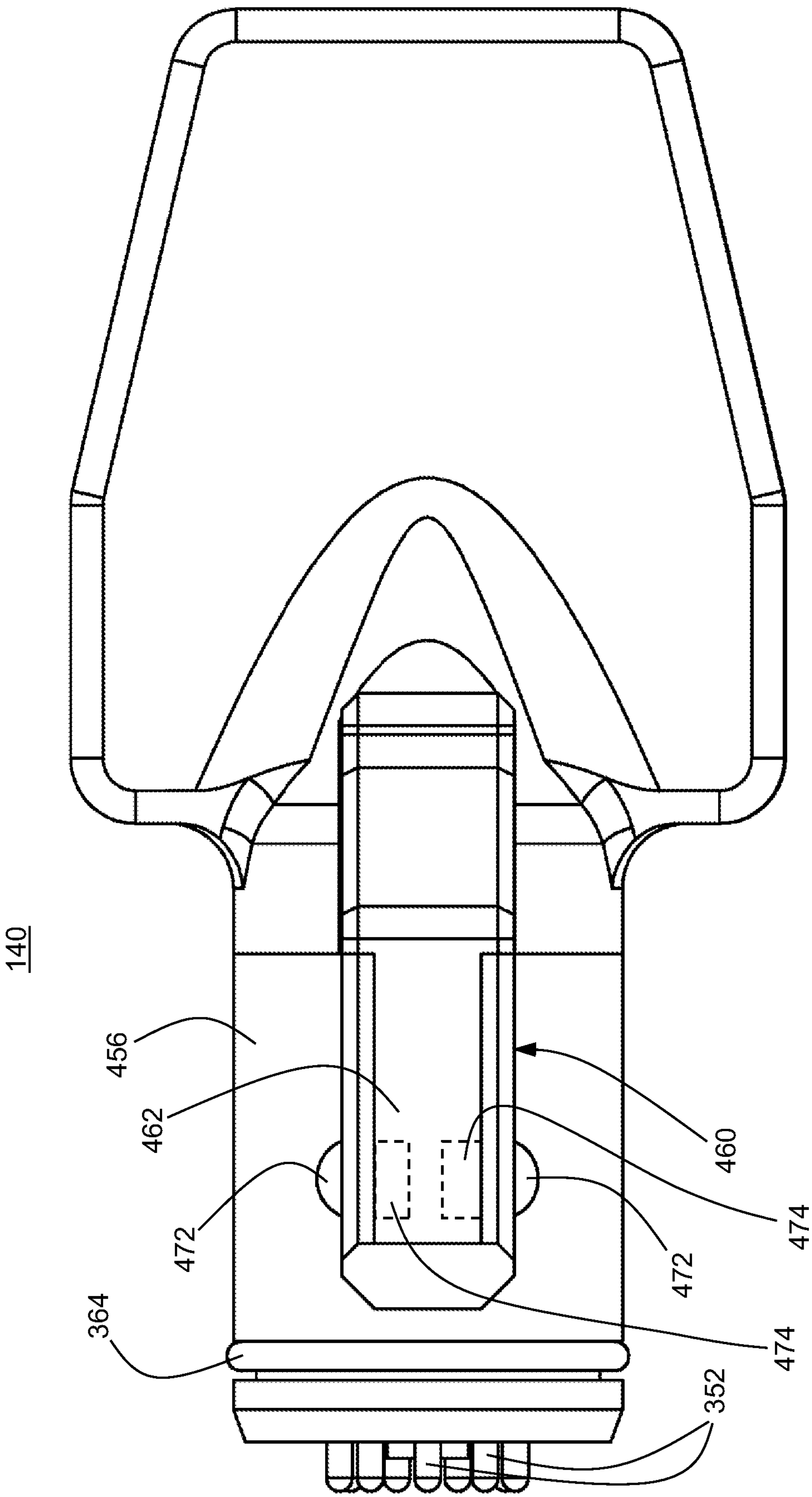


FIG. 13

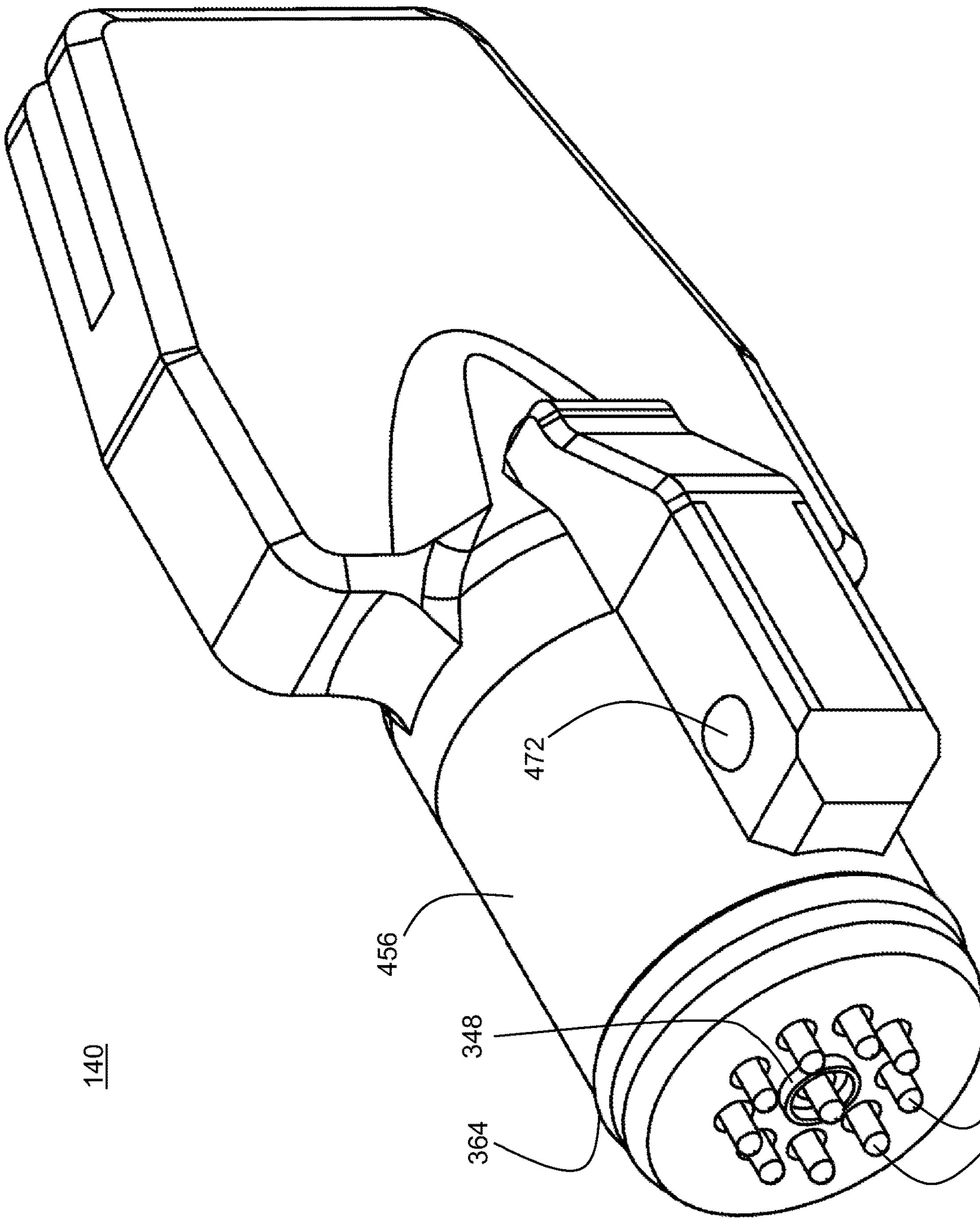


FIG. 14

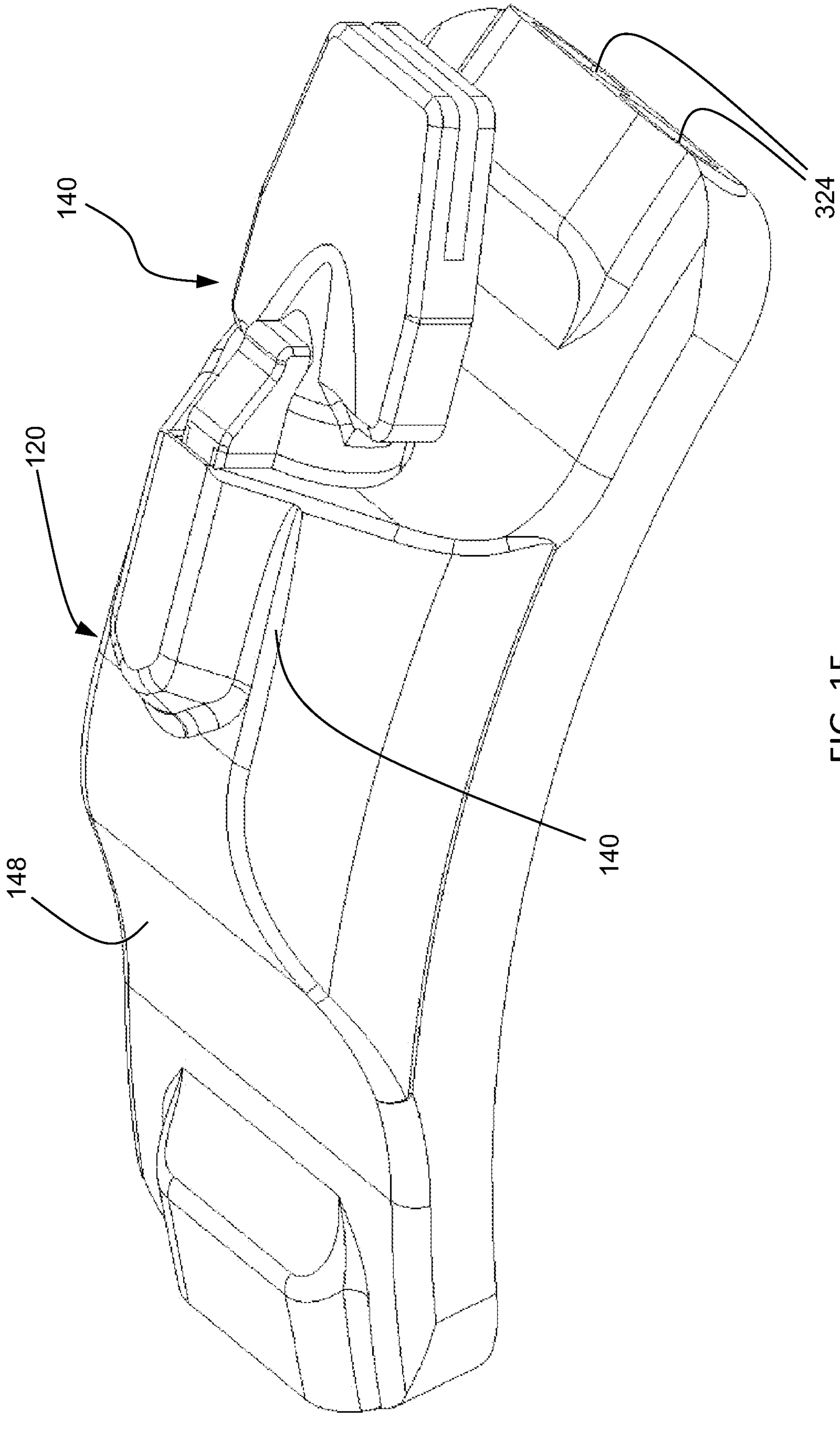


FIG. 15

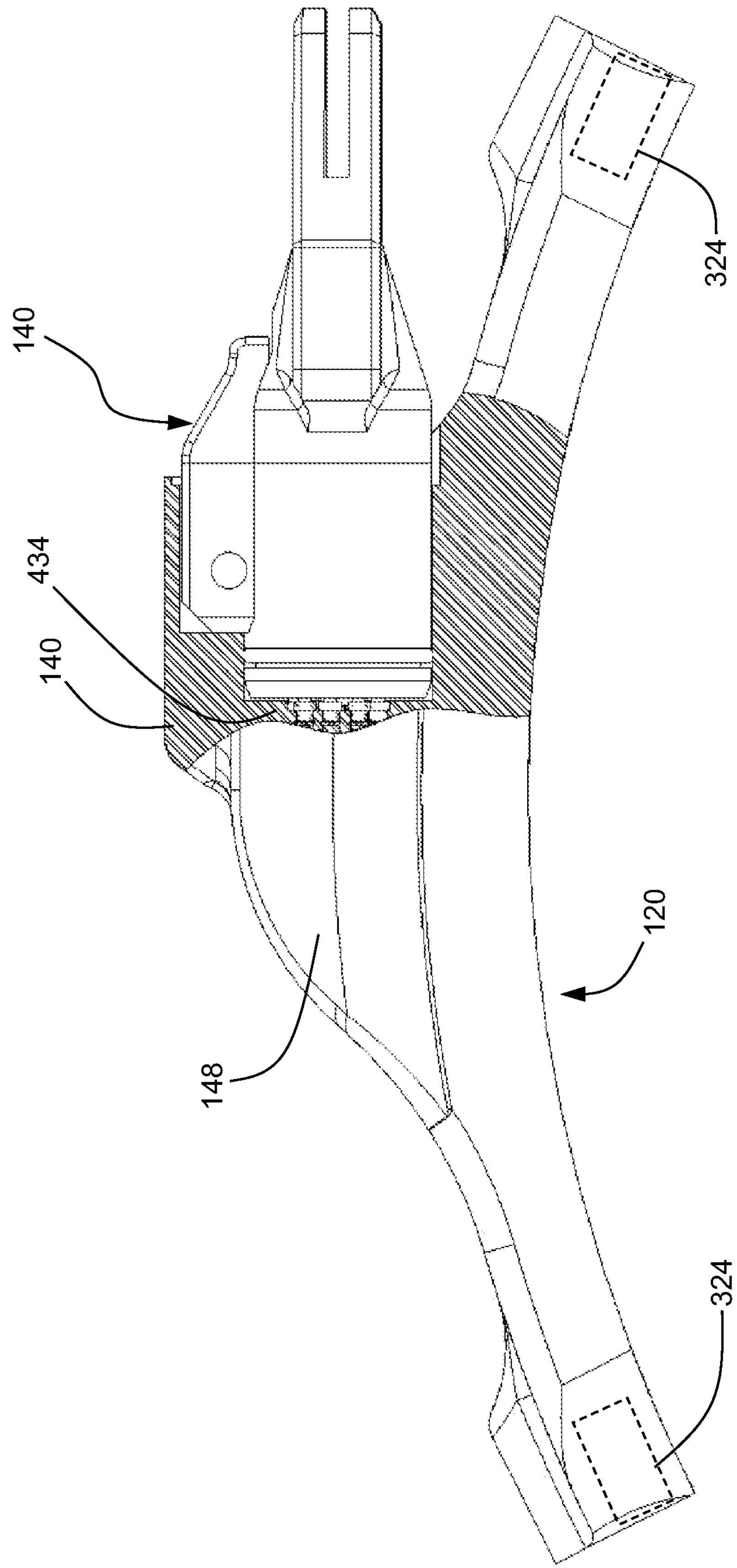


FIG. 16

WEAPON SLING WITH INTEGRAL CONNECTORS FOR WIRED COMMUNICATION AND/OR POWER TRANSFER

CROSS REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of U.S. provisional application No. 63/444,302 filed Feb. 9, 2023. The aforementioned application is incorporated herein by reference in its entirety.

BACKGROUND

The present invention relates to generally to weapon slings and, in particular, to weapon slings having integrated electrical conductors for communication of power, control, and/or data signals between a computer-based information handling system on a weapon and a power, control, and/or data hub worn on the body of a user.

Weapon slings are used with weapons, such as rifles, grenade launchers, rocket launchers, or the like. Typically, such slings are used to retain or secure the weapon to the user, allow the user to carry the weapon in a hands free manner, support the weapon while shooting, etc.

Weapons commonly include an accessory rail having electronics mounted to the rail or within the rail. Exemplary electronics include laser sights and illuminators, range finders, fire control systems, night vision devices, thermal and other cameras, flashlights, displays, computer systems, and so forth. Such electronic devices may require power from an external power supply. Such electronic devices may also send and/or receive data and/or control signals.

Other electronic devices may be attached to other wearable articles, such as a helmet and others. However, connecting electronics on the weapon with other electronics disposed on a helmet or other article worn by the user has proven difficult. Cable connections extending between a weapon and the user have a tendency to snag and are easily damaged. Therefore, a need remains for a robust communication connector system for a weapon that readily interconnects electronic accessory components on a weapon with electronic accessory components worn elsewhere on the body of the user.

The present disclosure contemplates a new and improved weapon sling which addresses the above-referenced problems and others.

SUMMARY

In one aspect, a weapon sling for supporting a weapon comprises a strap member having a first free end and a second free end opposite the first free end and defining a main branch extending between the first free end and the second free end, the strap member being formed of an elongate hollow flexible element. A loop branch is attached to the main branch, the loop branch being configured to be looped around at least a portion of a body of a user. A first connector assembly is secured to the first free end and a second connector assembly secured to the second free end. The first connector assembly is configured for detachable coupling to a weapon connector disposed on the weapon. The second connector assembly is configured for detachable coupling to a hub connector disposed on a hub worn on the body of the user. A plurality of electrical conductors extend within the hollow flexible element for electrically coupling a plurality of electrical contacts on the first connector assembly with a corresponding plurality of electrical contacts on the second connector assembly, the plurality of electrical conductors for providing communications between a host computer on the weapon and the hub.

In a more limited aspect, the loop branch has a first fixed end attached to the main branch proximal the first free end of the main branch and a second fixed end attached to the main branch proximal the second free end of the main branch.

In another more limited aspect, the loop branch has a first fixed end attached to the main branch proximal the first free end of the main branch and a third free end configured to be detachably coupled to the weapon.

In another more limited aspect, the first connector assembly includes a locking mechanism configured to provide a locking engagement between the first connector assembly and the weapon connector.

In another more limited aspect, the first connector assembly includes a breakaway mechanism configured to disengage the first connector assembly from the weapon connector when a pulling force on the first connector assembly exceeds a preselected threshold. In another more limited aspect, the breakaway mechanism comprises at least one ball detent captured within a receptacle on the first connector assembly, the at least one ball detent configured to mate with at least one recess within the weapon connector.

In another more limited aspect, the second connector assembly includes a breakaway mechanism configured to disengage the second connector assembly from the hub connector when a pulling force on the second connector assembly exceeds a preselected threshold. In another more limited aspect, the breakaway mechanism comprises at least one ball detent captured within a receptacle on the second connector assembly, the at least one ball detent configured to mate with at least one recess within the hub connector.

In another more limited aspect, the strap member is formed of an elongate hollow textile element.

In another more limited aspect, at least a portion of the elongate hollow textile element is stretchable.

In another more limited aspect, at least a portion of the plurality of electrical conductors is folded back on itself for accommodating stretching while maintaining an electrical connection.

In another more limited aspect, at least a portion of the elongate hollow textile element is formed of a nylon webbing material.

In another more limited aspect, the second connector assembly comprises a Universal Serial Bus (USB) connector.

In another more limited aspect, the weapon is provided in combination with the hub.

In another more limited aspect, the hub comprises a housing adapted to be attached to be a garment.

In another more limited aspect, the hub comprises a sleeve attached to the housing, wherein the sleeve is adapted to be adapted to be attached to be the garment.

In another more limited aspect, the sleeve is configured to be detachably attached to a shoulder strap of the garment.

In another more limited aspect, the sleeve is configured to be selectively attached to a left shoulder strap and a right shoulder strap of the garment.

In another more limited aspect, the garment is a vest selected from the group consisting of a plate carrier vest, tactical vest, load bearing or load carrying vest.

Advantages and benefits of the present invention will become apparent to those of ordinary skill in the art upon reading and understanding the following detailed description of the preferred embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

The invention may take form in various components and arrangements of components, and in various steps and arrangements of steps. The drawings are only for purposes of illustrating preferred embodiments and are not to be construed as limiting the invention.

FIG. **1** is an isometric view of a weapon interconnected with an associated plate carrier vest using a weapon sling in accordance with a first exemplary embodiment, taken generally from the front and left side. The first embodiment weapon sling assembly defines a single point sling in that it has one point of attachment to the weapon.

FIG. **2** is an isometric view of a weapon interconnected with an associated plate carrier vest using a weapon sling in accordance with the first exemplary embodiment, taken generally from the rear and left side.

FIG. **3** is left side view of a weapon interconnected with an associated plate carrier vest using a weapon sling in accordance with the first exemplary embodiment.

FIG. **4** is an enlarged view of the region **4** appearing in FIG. **3**.

FIG. **5** is a plan view of the first embodiment weapon sling appearing in FIG. **1**.

FIG. **6** is a plan view of a second embodiment weapon sling. The second embodiment weapon sling assembly defines a two point sling in that it has two points of attachment to the weapon.

FIG. **7** is a plan view of the first embodiment weapon sling having a first end coupled to a connector on the weapon and a second end coupled to an electrical connector on the associated plate carrier vest.

FIG. **8** is a plan view of the electrical connector on the associated plate carrier vest.

FIG. **9** is a plan view of the cable carrying portion of the weapon sling with the outer cover removed for illustration purposes.

FIG. **10** is an end view of the weapon sling connector end for attaching to the weapon.

FIG. **11** is a side view of the weapon sling connector end appearing in FIG. **10**.

FIG. **12** is an isometric view of the weapon sling connector end appearing in FIG. **10**.

FIG. **13** is a side view of the weapon sling connector end for attaching to the associated plate carrier vest.

FIG. **14** is an isometric view of the weapon sling connector end appearing in FIG. **13**.

FIG. **15** is an isometric view of the connector portion of the plate carrier vest with the weapon sling connector end inserted.

FIG. **16** is a side partial cutaway view of the connector portion of the plate carrier vest with the weapon sling connector end inserted.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Reference will now be made in detail to presently preferred embodiments of the invention, one or more examples of which are illustrated in the accompanying drawings. Each example is provided by way of explanation of the invention, not limitation of the invention, which may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present inventive concept in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting but rather to provide an understandable description of the present development. In fact, it will be apparent to those skilled in the art that modifications and variations can be made in the present invention without departing from the scope or spirit thereof. For instance, features illustrated or described as part of one embodiment may be used on another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

The terms "a" or "an," as used herein, are defined as one or more than one. The term "another," as used herein, is defined as at least a second or more. The terms "including" and/or "having" as used herein, are defined as comprising (i.e., open transition). The term "coupled" or "operatively coupled," as used herein, is defined as indirectly or directly connected.

As used in this application, the terms "front," "rear," "upper," "lower," "upwardly," "downwardly," "left," "right," and other orientation descriptors are intended to facilitate the description of the exemplary embodiment(s) of the present invention, and are not intended to limit the structure thereof to any particular position or orientation.

All numbers herein are assumed to be modified by the term "about," unless stated otherwise. The recitation of numerical ranges by endpoints includes all numbers subsumed within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

Unless specified otherwise, the term "proximal" as used herein, e.g., concerning the relative position of a fixed end or attachment point in relation to a free end, is not limited to direct adjacency and it does not exclude the possibility of there being a measurable distance between the fixed end or attachment point and the free end. Rather, unless specified otherwise, the term "proximal" is intended to signify the absence of intervening fixed ends or attachment points disposed between the specified fixed end or attachment point and the referenced free end.

Referring now to the drawings, FIGS. **1**-**5** and **7** illustrate a weapon sling assembly **100** in accordance with a first exemplary embodiment. The weapon sling assembly **100** comprises an elongate hollow flexible element **104** defining a strapping element, which is preferably a hollow textile element **104**. In embodiments, the hollow textile element **104** is formed of a flexible textile material. In embodiments, the elongate hollow textile element **104** is formed of a tubular braided material, tubular woven material, tubular knot material, or the like. In embodiments, the elongate hollow textile element **104** is formed from a folded web that is secured along a longitudinal seam, such as via stitching, adhesive, heat sealing, zipper, clasps, or other fastener(s). In certain embodiments, the elongate hollow textile element **104** is formed of a natural or synthetic textile material. In certain embodiments, the elongate hollow textile element **104** is formed of a hollow nylon webbing material.

The elongate hollow textile element **104** comprises a first free end **108** configured for detachable attachment to a weapon **112** and a second free end **116** configured for detachable attachment to a hub **120** intended to be wearable by a user. The elongate hollow textile element **104** includes a main branch **124** which extends between the first free end **108** and the second free end **116**. The hub **120** is preferably a Universal Serial Bus (USB) hub, although other hub protocols are contemplated, including IEEE 1394 and others.

The first free end **108** terminates in a weapon connector assembly **128** which detachably engages a complementary connector element **132** disposed on a receiver **136** of the weapon **112**. The second free end **116** terminates in a hub connector assembly **140** which detachably engages a complementary connector element **144** on the hub **120** which is adapted to be worn on the body of the user. In the depicted embodiment, the hub **120** includes a housing **148** configured to be attached to a garment **152** or other wearable element worn by the user.

In the illustrated preferred embodiment, the garment **152** is a vest, such as plate carrier vest, tactical vest, load bearing or load carrying vest, or the like. In embodiments, the garment **152** has front and rear panels **156**, **160**, respectively, which are adjoined by left and right shoulder straps **164**, **168**, respectively. In embodiments, the hub housing **148** is attached to a sleeve **172** slidably receiving the right shoulder strap **168** to allow positioning the hub **120** on the right shoulder of the user. Positioning the hub **120** on the left shoulder strap **164** to accommodate a left-handed shooter is also contemplated.

The elongate hollow textile element **104** further includes a loop branch **176**. The loop branch **176** has a first end **180** which diverges from the main branch **124** proximal the first free end **108** and a second end **184** which diverges from the main branch **124** proximal the second free end **116** to form a loop **188**. In embodiments, the junctions **180**, **184** between the main branch **124** and the loop branch **176** may be secured by stitching **178**. The loop branch **176** is secured to the main branch **124** to form a loop which passes over the right shoulder and under the user's left arm. In the illustrated embodiment, the loop branch **176** is secured to the main branch **124** to form a continuous loop. In alternative embodiments, the loop branch **176** may have free ends which are attachable together using fastener elements buckles or clasps (not shown) as would be understood by persons skilled in the art. In such alternative embodiments, such fastener elements may have adjustment features to adjust the size of the loop **188**, as would be understood by persons skilled in the art.

A portion of the elongate hollow textile element **104** main branch **124** between the junction **184** and the second free end **116** has a section **192** formed of an elastic or resiliently stretchable material to provide freedom of movement of the user without excess slack. Any electrical conductors, as described below, within the stretchable portion **192** of the main branch **124** may have portions that are folded back upon themselves within the sleeve **104** to accommodate stretching of the sleeve stretchable portion **192**. In certain embodiments, the stretchable portion **192** is able to stretch, e.g., from approximately 20% to 100% beyond its original, unstretched length.

Referring now to FIG. **6**, there is shown a weapon sling assembly **200** in accordance with a second exemplary embodiment. The weapon sling assembly **200** comprises an elongate hollow textile element **204** defining a strapping element. The hollow textile element **204** may be formed of the materials as described above with respect to the elongate hollow textile element **104**.

The elongate hollow textile element **204** comprises a first free end **208** configured for detachable attachment to the weapon **112** and a second free end **216** configured for detachable attachment to the hub **120**. The elongate hollow textile element **204** includes a main branch **224** which extends between the first free end **208** and the second free end **216**.

The first free end **208** terminates in a weapon connector assembly **228** which detachably engages the complementary connector element **132** disposed on a receiver **136** of the weapon **112**. The second free end **216** terminates in a hub connector assembly **240** which detachably engages the complementary connector element **144** on the hub **120**.

The elongate hollow textile element **204** further includes a loop branch **276**. The loop branch **276** has a first end **280** which diverges from the main branch **224** intermediate the first and second free ends **208**, **216**. The loop branch **276** has a second, free end **284** having a fastener element **286** for detachably attaching the free end **284** a complementary fastener element (not shown) on the weapon **112**. In embodiments, the complementary fastener element (not shown) is disposed on a buttstock portion **114** of the weapon. In embodiments, the fastener element **286** is a quick disconnect (QD) sling swivel as are generally known in the art.

The junction **280** between the main branch **224** and the loop branch **276** may be secured by stitching. In the illustrated embodiment, the loop branch **276** is of fixed length. In alternative embodiments, the loop branch **276** may further include an adjustment mechanism, such as a slide buckle or similar strap adjuster as would be understood by persons skilled in the art to adjust the length of the loop branch **276**.

A portion of the elongate hollow textile element **204** main branch **224** between the junction **284** and the second free end **216** has a section **292** formed of an elastic or resiliently stretchable material to provide freedom of movement of the user without excess slack. Any electrical conductors within the stretchable portion **292** of the main branch **224** may have portions that are folded back upon themselves within the sleeve **204** to accommodate stretching of the sleeve **204**.

As best seen in FIG. **7**, the weapon connector assembly **132** includes a first receptacle **300** for receiving the first connector end **128** and a second receptacle **304** for receiving a first connector end **308***a* of a cable **312**. In the illustrated embodiment, the cable **312** is a USB Type-C cable. A second connector end **308***b* of the cable **312** is connected to connection port **316** of a host computer system **320**, e.g., disposed on the weapon **112**. The computer system **320** is configured to function as a central controller for integrating multiple accessory devices associated with the weapon **112** and/or the user. In certain embodiments, the computer system **320** is a central network controller as described in commonly-owned U.S. Patent Application Nos. 63/423,324 filed Nov. 7, 2022, and Ser. No. 18/381,937 filed Oct. 19, 2023, each entitled "Central Network Controller for Weapon Accessory Devices," wherein the entire contents of each of the aforementioned applications, including the drawings, are incorporated herein by reference in their entireties. In certain embodiments, the connection ports **300** are locking USB Type-C connection ports as shown and described in the aforementioned U.S. Patent Application No. 63/423,324 and Ser. No. 18/381,937.

Referring now to FIG. **8**, and with continued reference to FIGS. **1-7**, the hub **120** includes a plurality of connection ports **324** configured to receive a first connection end **308***a* of a complementary cable **312**. In the illustrated embodiment, the hub **120** has four USB Type-C connection ports **324**, although other numbers of ports and hub protocols are contemplated. In certain embodiments, the connection ports **324** are locking USB Type-C connection ports as shown and described in the aforementioned U.S. Patent Application No. 63/423,324 and Ser. No. 18/381,937.

Referring now to FIG. **9**, and with continued reference to FIGS. **1-8**, there is illustrated the main branch **124** extending between the first free end **108** and the second free end **116** having one side of the elongate hollow textile element **104** removed for illustration purposes. The description below is equally applicable to the main branch **224** of the sling assembly **200**.

The main branch **124** includes a wide band antenna **328** which is electrically coupled to a wireless interface (RF transceiver) of the computer system **320** via a coaxial cable **332**. The internal cabling within the elongate hollow textile element **104** further includes three twisted conductors pairs **336** for data transfer, and power and ground conductors **340**, **344**, respectively, for power transfer.

Referring now to FIGS. **10-12**, and with continued reference to FIGS. **1-9**, the connector **128** includes a coaxial connector **348** which is electrically coupled to the coaxial cable **332**. A plurality of connector pins **352** (nine in the illustrated embodiment) align with complementary connector terminals or pads (not shown) disposed within the weapon connector receptacle **300** on the weapon connector assembly **132**. Each connector pin **352** is electrically coupled to a corresponding conductor within the main branch **124**, including the twisted pair conductors, power, and ground conductors. One or more of the pins **352** may also be coupled to a drain wire, metallic or conductive shield, e.g., for carrying unwanted electrical noise to ground. In embodiments, the connector pins **352** are spring-loaded telescoping (pogo) pins.

The connector assembly **128** includes a cylindrical body **356** with an axially extending projection **360**. A sealing ring **364** prevents entry of moisture or other external contamination. The projection **360** aligns with a complementary axial channel in the receptacle **300** of the connector assembly **132** and provides a keyed configuration to ensure proper alignment of the connector **128** within the receptacle **300**.

A ball detent locking feature allows for locking retention of the connector **128** within the receptacle **300**. The projection **360** defines a housing **362** enclosing an axially-extending channel which slidably receives a retractable plunger **368**. Captured ball members **372** extend through the projection housing **360** and are aligned with and engage complementary recesses or depressions within the receptacle **300** when the connector **128** is locked within the receptacle **300**.

A pull tab **376** is coupled to the plunger **368** to allow manual axial sliding movement of the plunger **368** within the housing **362**. When the plunger **368** is fully inserted into the housing **362**, the plunger **368** contacts the captured balls **372** to urge them outwardly and prevent them from moving inwardly to thereby lock the connector **128** within the receptacle **300**. When the pull tab **376** is pulled in the axial direction to retract the plunger **368** so that the plunger **368** disengage with the ball members **372**, the ball members **372** are free to move inwardly and disengage from the complementary recesses or depressions within the receptacle **300** to allow manual removal of the connector **128** from the receptacle **300**. In embodiments, a captured spring or other resiliently compressible member **374** within the housing **362** bears against the plunger **368** for biasing the plunger into the locked position. In this manner, disengagement between the connector end **128** and the weapon mounted connector **132** is prevented unless the tab **376** is pulled with sufficient force to overcome the bias of the spring element **374**, thereby reducing the chance of inadvertent disengagement of the connector end **128** from weapon mounted connector **132**.

Referring now to FIGS. **13** and **14**, and with continued reference to FIGS. **1-12**, the connector **116** includes a coaxial connector **348** which is electrically coupled to the coaxial cable **332**. A plurality of connector pins **352** align with complementary connector terminals or pads disposed within a connector receptacle **434** on the hub connector assembly **120**. Each connector pin **352** is electrically coupled to a corresponding conductor within the main branch **124**, including the twisted pair conductor, power, and ground conductors. One or more of the pins **352** may also be coupled to a drain wire, metallic or conductive shield, e.g., for carrying unwanted electrical noise to ground. In embodiments, the connector pins **352** are spring-loaded telescoping (pogo) pins.

The connector assembly **116** includes a cylindrical body **456** with an axially extending projection **460**. A sealing ring **364** prevents entry of moisture or other external contamination. The projection **460** aligns with a complementary axial channel in the receptacle **434** of the hub assembly **120** and provides a keyed configuration to ensure proper alignment of the connector **116** within the receptacle **434**.

A resilient ball detent breakaway retention mechanism allows for positive retention of the connector **116** within the receptacle **434**, wherein the connector **116** will disengage from the receptacle **434** upon application of a pulling force which overcomes the retention force of the resilient ball detent breakaway mechanism. The projection **460** defines a housing **462** enclosing an axially-extending channel which receives captured ball members **472** that extend through the projection housing **460**. The ball members **472** are aligned with and engage complementary recesses or depressions within the receptacle **434** when the connector **116** is received within the receptacle **434**. The ball members **472** are resiliently biased in the outward direction by one or more captured resiliently compressible members **474**, such as a coil spring or other resiliently compressible material.

Manually pulling the connector end **116** with sufficient force to compress the compressible/spring member(s) **474** allows causes the ball members **472** to disengage from the complementary receptacles within the receptacle **434** against the urging to the spring member(s) **474** to allow withdrawal of the connector end **116** from the receptacle **434**. In embodiments, as best seen in FIG. **9**, a like breakaway mechanism is provided on the connector **128**.

Referring now to FIGS. **15** and **16**, and with continued reference to FIGS. **1-14**, the hub **120** is operatively coupled to the host computer system **320** via the main branch portion **124** of the sling assembly **100**. The hub housing **148** further includes a plurality of connection ports **324** for coupling a plurality of electronically operated devices to the host computer system **320**. In certain embodiments, the hub **120** includes four connection ports **324**. In certain embodiments, the connection ports are USB Type-C connection ports. In certain embodiments, the connection ports **324** are locking USB Type-C connection ports as shown and described in the aforementioned U.S. Patent Application No. 63/423,324 and Ser. No. 18/381,937. In embodiments, the hub **120** is configured to be operatively coupled to a helmet mounding assembly, such as the helmet mounting assembly shown and described in commonly-owned U.S. Provisional Patent Application No. 63/427,496 filed Nov. 23, 2022, entitled "Helmet System," and U.S. patent application Ser. No. 18/500,657 filed Nov. 2, 2023, entitled "Helmet Accessory System," wherein the entire contents of each of the aforementioned applications, including the drawings, are incorporated herein by reference in their entireties.

The invention has been described with reference to the preferred embodiment. Modifications and alterations will occur to others upon a reading and understanding of the preceding detailed description. It is intended that the invention be construed as including all such modifications and alterations insofar as they come within the scope of the appended claims and their legal equivalents.

What is claimed is:

1. A weapon sling for supporting a weapon, comprising:

a strap member having a first free end and a second free end opposite the first free end and defining a main branch extending between the first free end and the second free end, the strap member being formed of an elongate hollow flexible element;

a loop branch attached to the main branch, the loop branch configured to be looped around at least a portion of a body of a user;

a first connector assembly secured to the first free end and a second connector assembly secured to the second free end;

the first connector assembly configured for detachable coupling to a weapon connector disposed on the weapon;

the second connector assembly configured for detachable coupling to a hub connector disposed on a hub worn on the body of the user;

a plurality of electrical conductors extending within the elongate hollow flexible element for electrically coupling a plurality of electrical contacts on the first connector assembly with a corresponding plurality of electrical contacts on the second connector assembly, the plurality of electrical conductors for providing communications between a host computer on the weapon and the hub.

2. The weapon sling of claim **1**, wherein the loop branch has a first fixed end attached to the main branch proximal the first free end of the main branch, and wherein the loop branch has a second fixed end attached to the main branch proximal the second free end of the main branch.

3. The weapon sling of claim **1**, wherein the loop branch has a first fixed end attached to the main branch proximal the first free end of the main branch, and wherein the loop branch has a third free end configured to be detachably coupled to the weapon.

4. The weapon sling of claim **1**, wherein the first connector assembly includes a locking mechanism configured to provide a locking engagement between the first connector assembly and the weapon connector.

5. The weapon sling of claim **1**, wherein the first connector assembly includes a breakaway mechanism configured to disengage the first connector assembly from the weapon connector when a pulling force on the first connector assembly exceeds a preselected threshold.

6. The weapon sling of claim **5**, wherein the breakaway mechanism comprises:

at least one ball detent captured within a receptacle on the first connector assembly, the at least one ball detent configured to mate with at least one recess within the weapon connector.

7. The weapon sling of claim **1**, wherein the second connector assembly includes a breakaway mechanism configured to disengage the second connector assembly from the hub connector when a pulling force on the second connector assembly exceeds a preselected threshold.

8. The weapon sling of claim **7**, wherein the breakaway mechanism comprises:

at least one ball detent captured within a receptacle on the second connector assembly, the at least one ball detent configured to mate with at least one recess within the hub connector.

9. The weapon sling of claim **1**, wherein the strap member is formed of an elongate hollow textile element.

10. The weapon sling of claim **9**, wherein at least a portion of the elongate hollow textile element is stretchable.

11. The weapon sling of claim **10**, wherein at least a portion of the plurality of electrical conductors is folded back on itself for accommodating stretching while maintaining an electrical connection.

12. The weapon sling of claim **1**, wherein at least a portion of the elongate hollow flexible element is formed of a nylon webbing material.

13. The weapon sling of claim **1**, wherein the second connector assembly comprises a Universal Serial Bus (USB) connector.

14. The weapon sling of claim **1**, in combination with said hub.

15. The weapon sling of claim **14**, wherein the hub comprises a housing adapted to be attached to be a garment.

16. The weapon sling of claim **15**, wherein the hub comprises a sleeve attached to the housing, wherein the sleeve is adapted to be adapted to be attached to be the garment.

17. The weapon sling of claim **16**, wherein the sleeve is configured to be detachably attached to a shoulder strap of the garment.

18. The weapon sling of claim **17**, wherein the sleeve is configured to be selectively attached to a left shoulder strap and a right shoulder strap of the garment.

19. The weapon sling of claim **17**, wherein the garment is a vest selected from the group consisting of a plate carrier vest, tactical vest, load bearing or load carrying vest.

* * * * *